Recitation 4: MIT 6.041 Probability Systems Analysis

1. n stage experiment:

1st person – 365 days, 2nd person – 364 days, ...,

nth person – 365 − n + 1 days

⇒ Probability = $\frac{365 \cdot 364 \cdot 363 \cdots (365-n+1)}{365^n}$

2. n stage experiment:

1st rook – 64 squares, 2nd rook – 49 squares,

3rd rook – 36 squares, ..., 8th rook – 1 square

Available squares:

1st rook – 64, 2nd rook – 63, 3rd rook – 62, ...,

8th rook – 57

⇒ Probability = $\frac{64 \cdot 49 \cdot 36 \cdot 25 \cdot 16 \cdot 9 \cdot 4 \cdot 1}{64 \cdot 63 \cdot 62 \cdot 61 \cdot 60 \cdot 59 \cdot 58 \cdot 57}$

These are not independent events because once a rook is placed, other rooks can occupy the remaining squares.

3. All possible outcomes = ${}^nC_k$

Favorable outcomes: 2 stages – draw i red balls from m red balls and the remaining k − i red balls from n − m balls = ${}^mC_i \cdot {}^{n-m}C_{k-i}$

$\Rightarrow$ Probability $= \dfrac{{}^{m}C_i \cdot {}^{n-m}C_{k-i}}{{}^{n}C_k}$

$= \dfrac{m!}{i!\,(m-i)!} \cdot \dfrac{(n-m)!}{(k-i)!\,(n-m-k+i)!} \cdot \dfrac{k!\,(n-k)!}{n!}$

4. Consider n objects, n_i of category i and $\sum_{i=1}^{r} n_i = n$

Total arrangements $= n!$

Repeating arrangements that lead to same overall arrangement $= n_1!\, n_2! \ldots n_r!$

$\Rightarrow$ Distinct arrangements $= \dfrac{n!}{n_1!\, n_2! \ldots n_r!}$

5. p_i = probability of drawing a ball with color i.

r stage experiment - each stage n_i draws.

Stage 1: probability of n_1 balls of color 1 $= {}^{n}C_{n_1}\, p_1^{n_1}$

Stage 2: probability of n_2 balls of color 2 $= {}^{n-n_1}C_{n_2}\, p_2^{n_2}$

...

Probability of n_i balls of color i in all stages

$= {}^{n}C_{n_1} \cdot {}^{n-n_1}C_{n_2} \cdot {}^{n-n_1-n_2}C_{n_3} \cdots {}^{n_r}C_{n_r} \prod_{i=1}^{r} p_i^{n_i}$

$= \dfrac{n!}{n_1!\, n_2! \cdots n_r!}\, p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r}$